TriPAF-Net: Adaptive Physical-Prior Fusion for Perception-Oriented Dehazing

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***Abstract*—Image restoration for autonomous driving should preserve task-relevant evidence rather than only minimize pixel error. We propose TriPAF-Net, a physics-guided dual-stream network that encodes dark-channel, bright-channel, and sky priors and fuses them with RGB features through five input-conditioned blocks using channel, spatial, haze-context, and high-frequency evidence. A parameter-matched control fixes both prior and detail gates at 0.5, while the decoder combines direct, learned-physical, and deterministic-prior reconstruction candidates. On 208 held-out CARLA pairs, TriPAF-Net achieves the highest precision (0.3903) and recall (0.1326) with a frozen YOLOv8m, although its PSNR (14.235 dB) is lower than both the fixed-gate model (14.626 dB) and foggy input (16.026 dB). On all 500 Foggy Cityscapes validation images at $\beta = 0.02$, it reaches 0.4381 mAP@0.50 and 0.2823 mAP@0.50:0.95, improving over foggy RGB by 0.0106 and 0.0131, respectively. The results show a restoration–perception mismatch and support input-conditioned physical-prior fusion for perception-oriented dehazing.**

Source code: https://github.com/LMJ-wakame/TriPAF-Net.

***Index Terms*—autonomous driving, image dehazing, physical priors, adaptive feature fusion, dark channel, bright channel, object detection, Foggy Cityscapes**

I. Introduction

Fog reduces contrast through attenuation and airlight, degrading structures important to driving perception, including vehicles, pedestrians, lanes, signals, and distant-object color. Strong dehazing may sharpen images but harm detector calibration, while milder enhancement can preserve task-relevant evidence despite lower pixel fidelity. DCP provides interpretable cues but is unreliable in bright and sky regions [1]; bright-channel evidence complements it in high-intensity areas [2], [3]. Deep models offer more flexible mappings [4]–[7], but simple prior concatenation is not input-adaptive, and better restoration does not necessarily improve detection [8], [9].

TriPAF-Net keeps three interpretable priors explicit and adapts their contributions to each input through:

- *Prior-as-representation:* DCP, BCP, and the sky mask are encoded by a dedicated branch for hierarchical feature learning.
- *Input-conditioned fusion:* channel, spatial, haze-context, and detail evidence control RGB–prior fusion at five scales, with a fixed-gate control isolating input dependence.
- *Restoration–perception evaluation:* one frozen detector is evaluated using native CARLA and converted Foggy Cityscapes annotations, retaining both negative pixel-fidelity and positive detection results.

We do not claim universal dehazing gains: CARLA pixel metrics favor other inputs. Instead, the results show that input-conditioned physical priors can change precision and recall, with the effect also observed on a public road benchmark.

II. Related Work

A. Physical and Learned Dehazing

The atmospheric scattering model underlies DCP and related methods [1], while bright-channel formulations complement dark statistics in difficult high-intensity regions [2], [3]. Learning-based approaches include AOD-Net for all-in-one inversion [4], GridDehazeNet and FFA-Net for multi-scale or attention-based feature aggregation [5], [6], and DehazeFormer for transformer-based restoration [7]. RESIDE provides a broad synthetic/real benchmark [10], and domain-adaptation methods explicitly address distribution shift [11].

B. Adverse-Weather Perception

Image-adaptive YOLO and TogetherNet connect enhancement with detection [8], [9]. Our setting differs in two ways: detector loss has zero weight during training, so detection is used only for downstream evaluation; and CARLA uses simulator-native boxes rather than teacher pseudo-labels. Cityscapes [12] and Foggy Cityscapes [13] further provide real urban geometry with synthesized depth-dependent fog for complementary evaluation.

III. Physical Evidence and Haze Context

A. Image Formation and Priors

For a pixel x, the atmospheric scattering model is

$$I(x) = J(x)t(x) + A[1 - t(x)], \qquad t(x) = e^{-\beta d(x)}, \quad (1)$$

where $I, J \in [0,1]^3$ denote foggy and clear radiance, $A \in [0,1]^3$ is atmospheric light, β is the scattering coefficient, and $d(x)$ is depth. For neighborhood $\Omega(x)$, the implemented prior definitions are

$$D(x) = \min_{y \in \Omega(x)} \min_{c \in \{r,g,b\}} I^c(y),$$
$$B(x) = \max_{y \in \Omega(x)} \max_{c \in \{r,g,b\}} I^c(y). \quad (2)$$

The deterministic sky estimate is

$$M_0(x) = \mathbf{1}[V(x) > 202,\ S_{\mathrm{HSV}}(x) < 50,\ H(x) < 50]\,,$$
$$M = \mathrm{Close}_{3\times3}\big(\mathrm{Open}_{3\times3}(M_0)\big)\,, \quad (3)$$

where OpenCV's 8-bit HSV convention is used and the morphology kernel is elliptical. The code uses a 5×5 rectangular neighborhood for D and B. All three maps are recomputed after paired spatial augmentation so that their pixels remain aligned with RGB.

B. Global Haze Descriptor

Let $\boldsymbol{y} = (0.299, 0.587, 0.114)^\top$, $Y(x) = \boldsymbol{y}^\top I(x)$, and let $\mu(\cdot)$ and $\sigma_0(\cdot)$ be the spatial mean and population standard deviation. The code constructs

$$\begin{aligned} \boldsymbol{h} &= [\mu(D), \mu(B), \mu(M), \mu(Y), \sigma_0(Y)] \in \mathbb{R}^5, \\ s &= \mathrm{sigmoid}(W_{s2}\,\mathrm{SiLU}(W_{s1}\boldsymbol{h}))\,, \qquad \boldsymbol{c} = [\boldsymbol{h}, s] \in \mathbb{R}^6. \end{aligned} \tag{4}$$

Here s is a learned scalar severity estimate. DCP, BCP, and M remain deterministic; only the mappings applied to their summary are learned.

IV. TriPAF-Net

A. Dual-Stream Multi-Scale Encoder

The RGB encoder widths are $(24, 48, 96, 192, 288)$, and the prior encoder widths are $(12, 24, 48, 96, 144)$. At level l, 1×1 projections place the two streams in the RGB feature space:

$$\begin{aligned} &\bar{X}_l = P_l^x X_l, \qquad \bar{P}_l = P_l^p P_l, \\ &R_l = \bar{X}_l - \mathrm{Avg}_{3\times3}(\bar{X}_l), \qquad E_l = \frac{1}{n_l}\sum_{c=1}^{n_l} |R_l^c|. \end{aligned} \tag{5}$$

R_l is a high-frequency residual, n_l is its channel count, and E_l is its channel-mean energy.

B. Input-Conditioned Prior and Detail Gates

Let $[\cdot,\cdot]$ denote channel concatenation and M_l the bilinearly resized sky map. The channel, spatial, and context terms are

$$\begin{aligned} C_l &= W_{l2}^c\,\mathrm{SiLU}\big(W_{l1}^c\,\mathrm{GAP}[\bar{X}_l, \bar{P}_l]\big)\,, \\ S_l &= \mathrm{Conv}_{5\times5}[\bar{X}_l, \bar{P}_l, M_l], \\ H_l &= W_{l2}^h\,\mathrm{SiLU}(W_{l1}^h \boldsymbol{c}). \end{aligned} \tag{6}$$

The scalar a_l controls the maximum departure from 0.5. The adaptive gates implemented in the code are

$$\begin{aligned} G_l^p &= \tfrac{1}{2} + \tfrac{1}{2}\tanh(a_l) \odot \tanh(C_l + S_l + H_l), \\ G_l^d &= \tfrac{1}{2} + \tfrac{1}{2}\tanh(a_l) \odot \tanh\big(\Psi_l[\bar{X}_l, \bar{P}_l, M_l, E_l, H_l]\big)\,. \end{aligned} \tag{7}$$

The learned amplitudes start at $a_l = 0.20$. Because $|\tanh(\cdot)| < 1$, each gate is bounded around 0.5, which prevents an unbounded switch at initialization. In TriPAF-Net Fixed, the executed intervention is exactly

$$G_l^p = G_l^d = 0.5, \tag{8}$$

while the remaining module structure is retained. The prior interpolation, nonlinear correction, and detail injection are

$$\begin{aligned} Q_l &= \bar{X}_l + G_l^p \odot (\bar{P}_l - \bar{X}_l), \\ \Delta_l &= \Phi_l[\bar{X}_l, \bar{P}_l, Q_l], \\ F_l &= X_l + \tanh(\gamma_l)\Delta_l + \tanh(\eta_l) G_l^d \odot R_l, \end{aligned} \tag{9}$$

with $\gamma_l = 0.10$ and $\eta_l = 0.05$ at initialization. Five fused features form U-Net-like skip connections to a four-stage decoder. Figure 1 separates learned and deterministic operations.

C. Learned Reconstruction Candidates

Let Z be the full-resolution decoder feature and Z_b the bottleneck. With $r_s = 0.5$ and $t_{\min} = 0.08$, the direct and learned-physical paths are

$$\begin{aligned} J_d &= \mathrm{clip}(I + r_s \tanh[R(Z)], 0, 1)\,, \\ \hat{t} &= t_{\min} + (1 - t_{\min})\,\mathrm{sigmoid}[T(Z)], \\ \hat{A} &= \mathrm{sigmoid}[A_h(Z_b)], \\ J_p &= \mathrm{clip}\left(\frac{I - \hat{A}}{\hat{t}} + \hat{A}, 0, 1\right). \end{aligned} \tag{10}$$

Define $E_I = \frac{1}{3}\sum_c |I^c - \mathrm{Avg}_{3\times3}(I^c)|$. A learned spatial mixture produces

$$\begin{aligned} q &= \mathrm{sigmoid}[B_h(Z) + \xi E_I], \\ J_r &= \mathrm{clip}(q \odot J_d + (1 - q) \odot J_p, 0, 1)\,, \end{aligned} \tag{11}$$

where ξ is learned. The internal deterministic-prior candidate estimates atmospheric colors from the top 0.1% pixels of D and B. Let these means be A_D, A_B, and $M_{15} = \mathrm{Avg}_{15\times15}(M)$. The exact reconstruction is

$$\begin{aligned} t_D &= 1 - 0.88D / \max(\min_c A_D^c, 0.05), \\ t_B &= |B - \max_c A_B^c| / \max(1 - \max_c A_B^c, 0.05), \\ t_\pi &= \mathrm{clip}((1 - M_{15})t_D + M_{15} t_B, 0.08, 1)\,, \\ A_\pi &= (1 - M_{15})A_D + M_{15} A_B, \\ J_\pi &= \mathrm{clip}\left(\frac{I - A_\pi}{t_\pi} + A_\pi, 0, 1\right). \end{aligned} \tag{12}$$

The training output combines the input, learned reconstruction, and prior candidate using image-wise softmax weights:

$$\begin{aligned} \boldsymbol{w} &= \mathrm{softmax}(W_o[\mathrm{GAP}(Z), \boldsymbol{c}])\,, \qquad \sum_{i=0}^{2} w_i = 1, \\ J_{\mathrm{raw}} &= \mathrm{clip}(w_0 I + w_1 J_r + w_2 J_\pi, 0, 1)\,. \end{aligned} \tag{13}$$

The public output key "image" is J_{raw} and is the quantity supervised by the active training loss.

D. Bounded Inference Stabilization

The evaluated stabilized output is not identical to J_{raw}. First, only w_0, w_1 are renormalized:

$$\begin{aligned} \bar{w}_i &= w_i / \max(w_0 + w_1, 10^{-6}), \quad i \in \{0, 1\}, \\ \lambda &= 0.65 + 0.25\bar{w}_1, \qquad N = I + \lambda(J_r - I), \\ r &= 0.22 \tanh[(N - I)/0.22], \\ r_Y &= \boldsymbol{y}^\top r, \qquad r_C = r - r_Y. \end{aligned} \tag{14}$$

With $M_{31} = \mathrm{clip}\big(\mathrm{Avg}_{31\times31}(M), 0, 1\big)$, $d_I = I - \mathrm{Avg}_{5\times5}(I)$, and

$$e = \left[1 + 24\,\mathrm{Avg}_{5\times5}\left(\tfrac{1}{3}\sum_c |d_I^c|\right)\right]^{-1}, \tag{15}$$

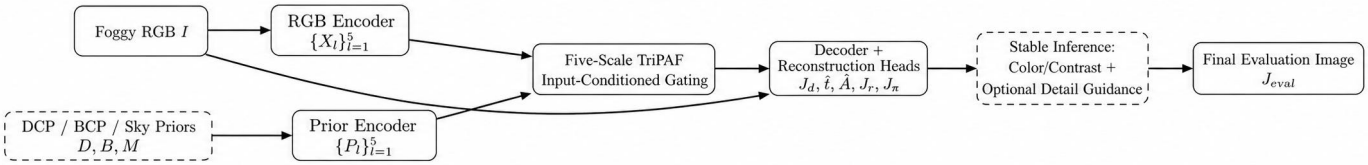


Fig. 1. TriPAF-Net architecture. Solid boxes denote learned modules; dashed boxes denote deterministic prior extraction and stable refinement.

the stabilized image before color refinement is

$$S_0 = I + e\big[(1 - 0.10M_{31})r_Y + 0.55(1 - 0.60M_{31})r_C\big],$$
$$S_1 = \mathrm{clip}\big(S_0 + 0.40[d_I - (S_0 - \mathrm{Avg}_{5\times5}\, S_0)], 0, 1\big). \quad (16)$$

Thus the learned correction is softly bounded, chroma is more strongly suppressed in sky regions, and a fraction of input detail is restored.

E. Stable Evaluation Output

After S_1, the implementation applies sky-protected luminance/chroma refinement and optional deterministic DCP/BCP luminance guidance. If S_2 is the bounded color-refined image and L_g, L_{S_2} are the guided and current luminance, the evaluated image is

$$J_{\mathrm{eval}} = \mathrm{clip}(S_2 + (L_g - L_{S_2}), 0, 1). \quad (17)$$

This stage changes luminance while retaining the learned chroma direction. Its coefficients are fixed; therefore the final image is not attributed solely to the adaptive gates. The complete code-level expansion is retained in the separate formula guide.

V. Training Objective

For prediction $P = J_{\mathrm{raw}}$ and clean target J, define

$$\mathcal{L}_{\mathrm{char}} = \frac{1}{N}\sum_i \sqrt{(P_i - J_i)^2 + 10^{-6}},$$
$$\mathcal{L}_{\nabla} = \|\nabla_x P - \nabla_x J\|_1 + \|\nabla_y P - \nabla_y J\|_1,$$
$$u = \mathrm{clip}\left(\frac{f - 20}{30}, 0, 1\right), \quad (18)$$

where f is CARLA fog density. The SSIM loss is

$$\mathrm{SSIM}(P, J) = \frac{(2\mu_P\mu_J + C_1)(2\sigma_{PJ} + C_2)}{(\mu_P^2 + \mu_J^2 + C_1)(\sigma_P^2 + \sigma_J^2 + C_2)},$$
$$\mathcal{L}_{\mathrm{SSIM}} = \mathrm{clip}(1 - \mathrm{mean}(\mathrm{SSIM}), 0, 2). \quad (19)$$

The code uses 3×3 average statistics, $C_1 = 0.01^2$, and $C_2 = 0.03^2$. Using SmoothL1 ℓ_{H}, the active objective is

$$\mathcal{L} = \mathcal{L}_{\mathrm{char}} + 0.2\mathcal{L}_{\mathrm{SSIM}} + 0.1\mathcal{L}_{\nabla} + 0.2\ell_{\mathrm{H}}(s, u)$$
$$+ 0.15\frac{1}{N}\sum_i (1 - u)^2 |P_i - I_i|. \quad (20)$$

The last term discourages unnecessary change in lighter fog. Detector loss is zero in the predefined configurations. The validation-only checkpoint score is

$$\mathcal{Q} = \overline{\mathrm{PSNR}} - 10\overline{\mathrm{MAE}} + 0.25(\overline{\mathrm{PSNR}}_{f\geq30} - \overline{\mathrm{PSNR}}), \quad (21)$$

because the configured detector term is inactive.

VI. Experimental Protocol

A. Datasets and Isolation

CARLA [14] supplies 1,436 synchronized foggy/clear pairs at 1024×1024, with fog density from 20 to 50. The predefined split contains 996 training, 232 validation, and 208 test pairs with zero overlap. Only the full 208-image test split is used in the CARLA tables. Foggy Cityscapes is evaluated at $\beta = 0.02$. Its train/validation partitions contain 2,975/500 images; the reported experiment uses all 500 validation images. gtFine polygons are converted to YOLO boxes for car, person, rider, truck, bus, train, motorcycle, and bicycle. Seven classes map directly to COCO; rider is excluded from AP aggregation rather than being forced into person.

B. Optimization and Detection

Both adaptive and fixed models use 20 epochs, seed 42, 512×512 paired crops, batch size 2, two-step gradient accumulation, AdamW at 2×10^{-4} with weight decay 10^{-4}, three warm-up epochs, cosine decay, mixed precision, gradient-norm cap 1.0, and EMA 0.999. Detection uses one frozen COCO-pretrained YOLOv8m [15]; no downstream fine-tuning is performed. For Foggy Cityscapes, detector input is 1024, confidence threshold 0.001, NMS IoU 0.7, and batch size 2.

C. Metrics

For N scalar RGB samples with unit dynamic range, let $\hat{J} = J_{\mathrm{eval}}$. Then

$$\mathrm{MSE} = N^{-1}\sum_i (\hat{J}_i - J_i)^2, \quad \mathrm{PSNR} = 10\log_{10}(1/\mathrm{MSE}),$$
$$\mathrm{MAE} = N^{-1}\sum_i |\hat{J}_i - J_i|. \quad (22)$$

CIEDE2000 is computed after conversion to CIE $L^*a^*b^*$ and reports the standard perceptual color difference. For detector counts TP, FP, and FN,

$$\mathrm{Precision} = \frac{TP}{TP + FP}, \qquad \mathrm{Recall} = \frac{TP}{TP + FN},$$
$$\mathrm{AP} = \int_0^1 p_{\mathrm{interp}}(r)\, dr. \quad (23)$$

mAP averages AP over classes; mAP@0.50 uses IoU 0.50, mAP@0.50:0.95 averages ten thresholds from 0.50 to 0.95.

TABLE I
RESTORATION MEANS ON 208 CARLA PAIRS.

Method	N	PSNR↑	RGB MAE↓	CIEDE↓
Foggy input	208	**16.026**	**0.11879**	**10.8999**
DCP/BCP	208	14.397	0.16934	15.5854
TriPAF-Net Fixed	208	14.626	0.16575	14.9174
TriPAF-Net	208	14.235	0.17335	15.6287

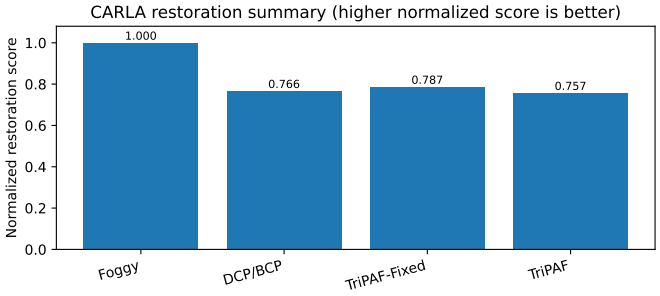


Fig. 2. Normalized CARLA restoration summary using the foggy input as the unit baseline. The normalized score is the mean of $\mathrm{PSNR}/\mathrm{PSNR}_{\mathrm{fog}}$, $\mathrm{MAE}_{\mathrm{fog}}/\mathrm{MAE}$, and $\mathrm{CIEDE}_{\mathrm{fog}}/\mathrm{CIEDE}$.

VII. RESULTS

A. CARLA Restoration and Detection

Table I reports a negative restoration result. The foggy input is best on all three pixel/color metrics. Fixed gates outperform adaptive gates by 0.391 dB PSNR and 0.00760 MAE. These values do not support a claim of superior paired reconstruction.

TABLE II
FROZEN YOLOV8M WITH CARLA-NATIVE ANNOTATIONS ON 208 IMAGES.

Input	Precision	Recall	mAP@0.50	mAP@0.50:0.95
Clear	0.3354	0.1253	0.0980	**0.0558**
Foggy	0.3341	0.1149	0.0948	0.0506
DCP/BCP	0.2700	0.1175	**0.1018**	0.0548
TriPAF-Net Fixed	0.3736	0.1127	0.0996	0.0545
TriPAF-Net	**0.3903**	**0.1326**	0.1013	0.0550

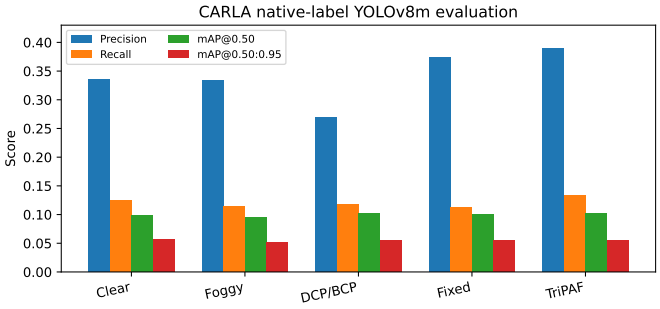


Fig. 3. CARLA native-ground-truth detection under the stated protocol. The adaptive output has the highest precision and recall, whereas DCP/BCP has the highest mAP@0.50.

The adaptive model raises precision by 0.0562 and recall by 0.0177 relative to foggy RGB. Its mAP@0.50 is 0.0065 higher than foggy RGB and only 0.0005 below DCP/BCP. The defensible interpretation is improved precision/recall balance, not universal dominance of every detection metric.



Fig. 4. Three CARLA test examples at different fog densities. Columns show the clear reference, hazy input, DCP/BCP, TriPAF-Net Fixed, and TriPAF-Net.

B. Foggy Cityscapes Cross-Dataset Test

TABLE III
FOGGY CITYSCAPES VALIDATION, $\beta = 0.02$, ALL 500 IMAGES.

Input	Precision	Recall	mAP@0.50	mAP@0.50:0.95
Foggy original	0.6231	0.3933	0.4275	0.2692
DCP/BCP	0.6534	0.4090	0.4329	0.2806
TriPAF-Net	**0.6571**	**0.4105**	**0.4381**	**0.2823**

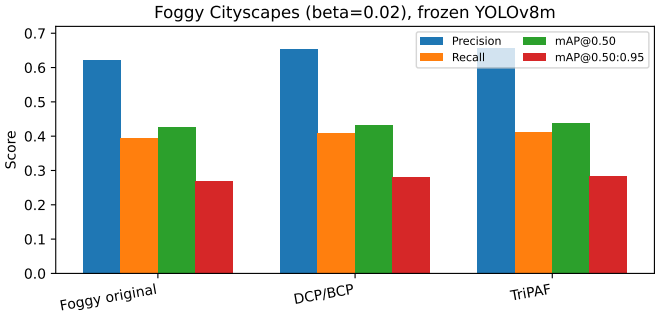


Fig. 5. Foggy Cityscapes detection using identical YOLOv8m settings for the three image sources.

On Foggy Cityscapes, TriPAF-Net improves every reported detector metric. Against foggy RGB, the absolute gains are 0.0340 precision, 0.0172 recall, 0.0106 mAP@0.50, and 0.0131 mAP@0.50:0.95. Against DCP/BCP, the gains are smaller: 0.0037, 0.0015, 0.0052, and 0.0017. Because the detector is frozen and the model was not trained with detector loss, this experiment is evidence of favorable input transformation under a distribution shift, not detector-specific training or statistical superiority.

C. Efficiency and Qualitative Behavior

The model has 13.23M parameters at all tested resolutions. Only the 384×384 configuration exceeds 30 FPS. Scaling from

TABLE IV
EFFICIENCY ON RTX 3060 TI, BATCH 1, FP16.

Input	GFLOPs	ms	FPS	Peak MB
384^2	82.7	29.63	**33.75**	247
512^2	147.0	47.83	20.91	370
640^2	229.8	73.04	13.69	534
1024^2	588.2	177.43	5.64	1225

384×384 to 1024×1024 increases GFLOPs, latency, and peak GPU memory by approximately $7.11\times$, $5.99\times$, and $4.96\times$, respectively, while reducing FPS by 83.3%. High-resolution operation is therefore an offline setting on the tested GPU.

VIII. DESIGN ANALYSIS AND EVIDENCE

The principal innovation is not the mere presence of three priors. It is the way their roles are made conditional at every scale. Equation (7) couples local channel/spatial evidence with the same global haze descriptor across the feature pyramid, while Eq. (9) retains the RGB stream as an explicit residual reference. This design gives a clear intervention: Eq. (8) removes input dependence without replacing the encoder or decoder. The second contribution is the explicit separation of representation, reconstruction, and evaluated output. Equations (13) and (17) show that the supervised image and final image differ; disclosing this boundary prevents deterministic refinement from being misattributed to learned fusion. The experiments also reveal that the meaning of better depends on the measurement. CARLA pixel metrics favor the unprocessed foggy image, CARLA mAP@0.50 favors DCP/BCP, and CARLA precision/recall favor adaptive fusion. On Foggy Cityscapes, adaptive fusion leads all four reported detection metrics. Together, these results motivate task-aware evaluation while warning against using one detector result to erase a negative restoration result.

IX. LIMITATIONS

Only seed 42 is reported, and Foggy Cityscapes still uses synthetic fog on real geometry. Multi-seed paired intervals, real-fog data, and a no-guidance factorial test are required before assigning the final gain independently to adaptive gates. Detection also depends on class mapping, detector weights, and thresholds.

X. CONCLUSION

TriPAF-Net treats dark, bright, and sky priors as a parallel representation and uses five input-conditioned fusion blocks to decide how they interact with RGB features. Code-aligned formulas show a bounded adaptive gate, hybrid reconstruction, learned-severity-conditioned refinement, and optional deterministic guidance. Experiments show a clear restoration–perception trade-off: adaptive fusion is not the strongest CARLA restorer, yet it improves CARLA precision/recall and leads all reported Foggy Cityscapes detection metrics. This evidence supports physically structured, input-conditioned fusion as a promising perception-oriented mechanism while leaving multi-seed, real-fog, and factorial validation as necessary next steps.

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